



DEPARTMENT OF COMPUTER ENGINEERING & APPLICATIONS

Institute of Engineering & Technology

Lab Manual

Subject Name & Code: Digital Image Processing Lab (BCSE0131)

Course: B. Tech.(CS)

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Section: D

Roll no.: 33

Year: III

Semester:

V

EXPERIMENT NO: 1

Objective: The objective of this lab session is to understand the image Arithmetic Operations & image Logical Operation and implement the following tasks in MATLAB.

1. Image addition
2. Image complement
3. Logical operations such as NOT, OR, AND, and XOR on images
4. Understand the usefulness of logical operations

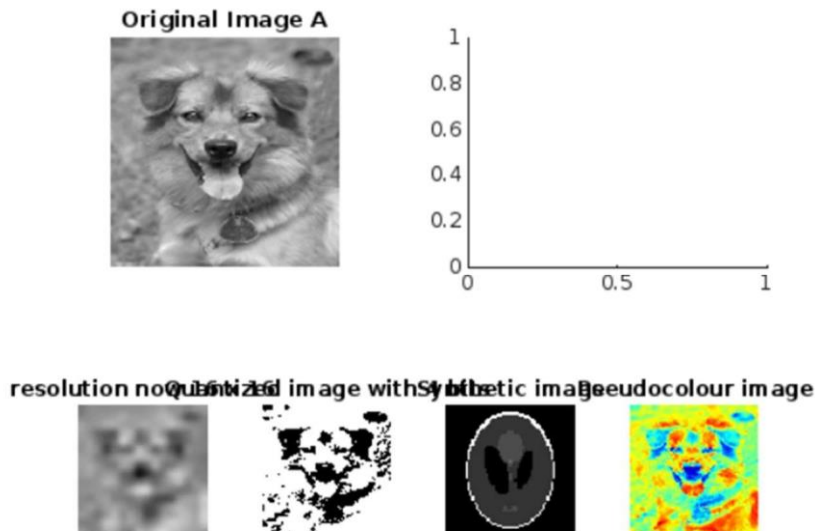
Perform Arithmetic operation (images Additions, Image Complements)

```
mygrayimg = imread('happy_dog.jpg'); scaledimageA =  
imresize(mygrayimg,[256,256]); subplot(2, 2, 1);  
imshow(scaledimageA,[]); title('Original Image A');
```

```
subplot(2, 2, 2); mygrayimg1 =  
imread('cb.jpg');  
scaledimageB = imresize(mygrayimg1,[256,256]); imshow(scaledimageB,[]); title('Original  
Image B');
```

```
d = imcomplement(scaledimageB); subplot(2, 2, 3);  
imshow(d,[]); title('Image complement');
```

```
modbrightimg = imadd(scaledimageA,50); subplot(2, 2, 4);  
imshow(modbrightimg); title('Add Brightness of the Image');
```



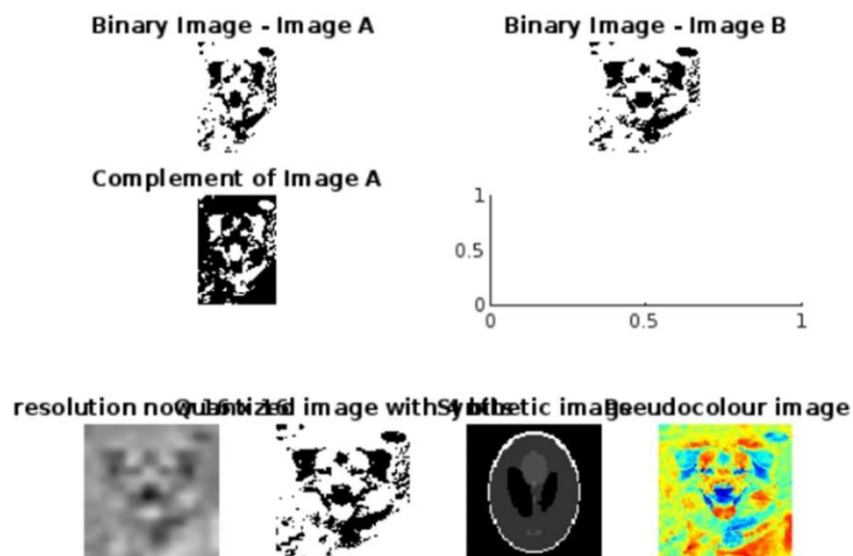
Perform Logical Operation on Images (NOT,OR,AND and XOR)

```

myimageA=imread('happy_dog.jpg'); mybinaryimageA =
im2bw(myimageA);

myimageB=imread('happy_dog.jpg'); myimageadjustB
=imresize(myimageB,[256,256]); mybinaryimageB =
im2bw(myimageadjustB);
subplot(4,2,1)
imshow(mybinaryimageA),title('Binary Image - Image A ');
subplot(4,2,2)
imshow(mybinaryimageB),title('Binary Image - Image B ');
subplot(4,2,3) resultor=
~mybinaryimageA ;
imshow(mat2gray(resultor)), title('Complement of Image A ');
subplot(4,2,4)
resultxor= xor(mybinaryimageA,mybinaryimageB); imshow(mat2gray(resultxor)), title('Image A
XOR Image B ');
subplot(4,2,5)
resultor= mybinaryimageA | mybinaryimageB;
imshow(mat2gray(resultor)), title('Image A OR Image B ');
subplot(4,2,6)
resultand= mybinaryimageA & mybinaryimageB;
imshow(mat2gray(resultand)), title('Image A AND Image B '); subplot(4,2,7)
mycolourimg = imread('zoo.jpeg'); mygrayimg =
rgb2gray(mycolourimg); subimage(mygrayimg),title('Original
Grey Image'); axis off; axis equal;
subplot(4,2,8) planeimg =
double(mygrayimg); k = 128;
mysevendgreyimg = mod(floor(planeimg/k),2); imshow(mysevendgreyimg),title('Seventh Plane');

```



EXPERIMENT NO: 2

Objective: Perform various Histogram Operations histogram plot, histogram Equalization, Contrast Stretching & gamma correction on images & plot histogram without using imhist function.

Performs various Histogram Operations (histogram plot, histogram Equalization, Contrast Stretching & gamma correction on images)

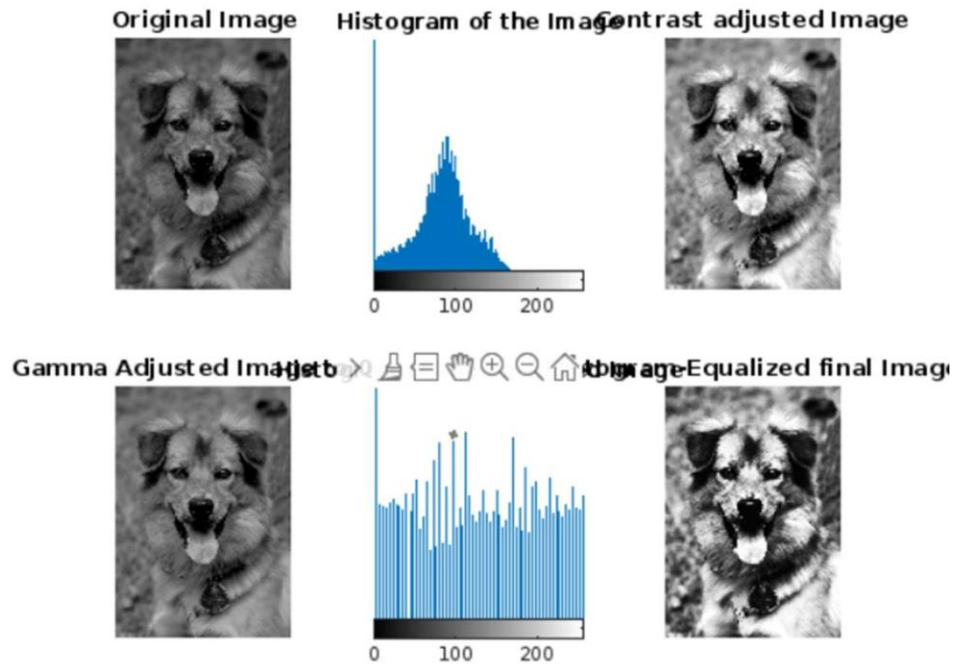
```
mycolourimage = imread('happy_dog.jpg');
mygrayimage = rgb2gray(mycolourimage);
mygrayimage = imsubtract(mygrayimage,60);
subplot(2,3,1); imshow(mygrayimage); title('Original
Image');

myimage = uint8(mygrayimage);
subplot(2,3,2); imhist(myimage); axis
off, axis tight;
title('Histogram of the Image');

subplot(2,3,3);
myadjustedimage = imadjust(myimage,stretchlim(myimage),[]);
imshow(myadjustedimage); title('Contrast adjusted Image');

subplot(2,3,4);
myadjustedgammaimage = imadjust(myimage,[],[],0.8);
imshow(myadjustedgammaimage); title('Gamma Adjusted Image to
0.8');

subplot(2,3,5);
[equalisedimage, T] = histeq(myimage);
imhist(equalisedimage); axis off, axis tight;
title('Histogram of the Equalized Image');
    subplot(2,3,6);
imshow(equalisedimage);
title('Histogram-Equalized final Image');
```

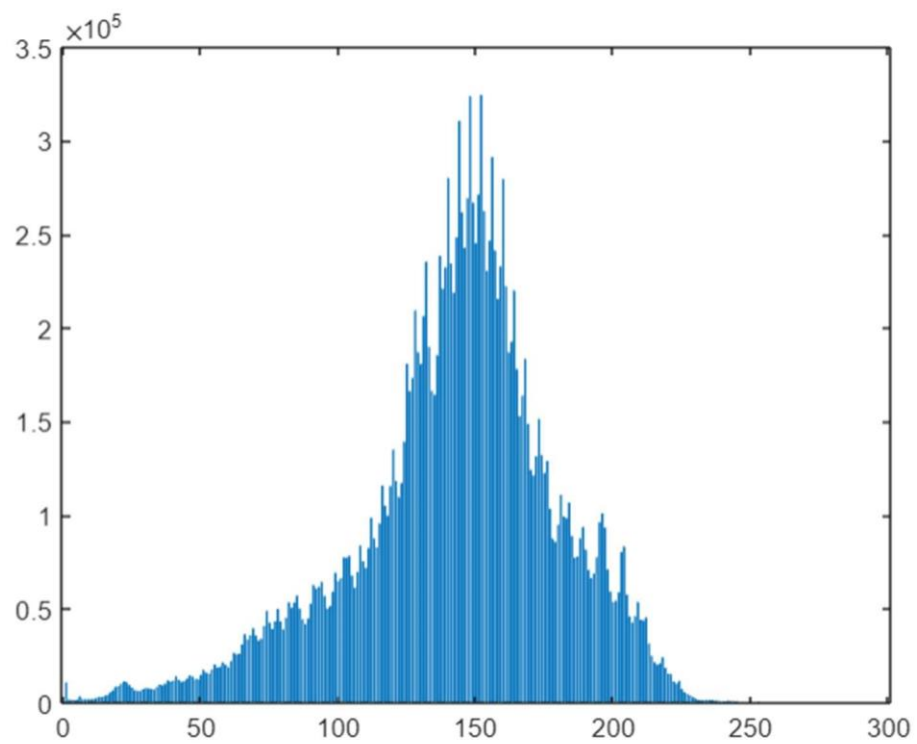


plot histogram without using imhist function

```

a=imread('happy_dog.jpg');
a=double(a); [row col]=size(a);
h=zeros(1,300); for n=1:1:row
for m=1:1:col    if a(n,m)==0
a(n,m)=1;      end  end end for
n=1:1:row    for m=1:1:col
t=a(n,m);    h(t)=h(t)+1;
end end figure(1)
imshow(uint8(a)); figure(2) bar(h)

```



EXPERIMENT NO: 3

Objective: Perform smoothing using linear and order statistics filters min, max & med of varying sizes and Sharpen an image using Laplacian filter.

Perform smoothing using linear and order statistics filters of varying sizes.

```
img2=imread('noise.jpeg');
img1=img2; [m,n]= size(img2);
size1 = input('Enter the size of filter : '); f = ones(size1); img2
= double(img2); img3 = double(img2); for i=2:m-1 for
j=2:n-1 s=0; for k=1:size1 for l=1:size1
s=s+img2(i-2+1, j-2+1)*f(k,l); end end
img3(i,j) = s/(size1^2); end end
subplot(1,2,1), imshow(img1); subplot(1,2,2),imshow(uint8(img3));
```



Perform smoothing image using Max Min & Med filtering i1 =
imread('dog.jfif'); i1 = rgb2gray(i1); [m,n] = size(i1); i1 = double(i1);
size1 = input('size of filter (odd number): '); f = ones(size1); c =
(size1+1)/2; maxi = i1; mini = i1; medi = i1; for i=c:m-c+1
for j=c:n-c+1
maxi(i,j) = max(max(i1(i-c+1:i-c+size1,j-c+1:j-c+size1))); mini(i,j) = min(min(i1(i-
c+1:i-c+size1,j-c+1:j-c+size1))); medi(i,j) = median(median(i1(i-c+1:i-c+size1,j-c+1:j-
c+size1))); end end
figure(1), subplot(2,2,1),imshow(uint8(i1)),title('Original'); figure(1),
subplot(2,2,2),imshow(uint8(maxi)),title('Max Filtered'); figure(1),
subplot(2,2,3),imshow(uint8(mini)),title('Min Filtered'); figure(1),
subplot(2,2,4),imshow(uint8(medi)),title('Median Filtered');

Original



Max Filtered



Min Filtered



Median Filtered



Sharpen an image using Laplacian filter.

```

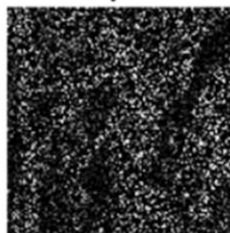
i1 = imread('lennaimage.jpg');
[m n] = size(i1); i1 = double(i1); f
= [0 1 0;1 -4 1; 0 1 0]; s = i1; for i
= 2:m-1 for j = 2:n-1 sum =
0; for k = 1:3 for l =
1:3
sum = sum + i1(i-2+k, j-2+l)*f(k,l); end end
s(i,j) = sum; end end
sm = i1 - s;
figure(1), subplot(1,3,1), imshow(uint8(i1)), title('Original'); figure(1), subplot(1,3,2),
imshow(uint8(s)), title('Sharpened'); figure(1), subplot(1,3,3), imshow(uint8(sm)),
title('Subtracted');

```

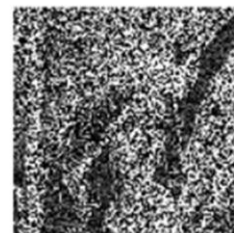
Original



Sharpened



Subtracted



EXPERIMENT NO: 4

Objective: The objective of this lab session is to understand and implement the Fast Fourier transform (FFT) and frequency domain filtering on images using MATLAB.

FFT on image

Low- pass filtered Image

Band-pass filtered image

```
mygrayimg = imread('dog.tif');  
mygrayimg = imresize(rgb2gray(mygrayimg),[256 256]); subplot(2,2,1);  
imshow(mygrayimg),title('Original Image');
```

```
myfftimage = fft2(mygrayimg); tmp =  
abs(myfftimage); mylogimg =  
log(1+tmp); subplot(2,2,2);  
imshow(mat2gray(mylogimg)); title('FFT  
Image');
```

```
[M,N] = size(myfftimage);  
low = 62;  
band1 = 15;  
band2 = 60;
```

```
mylowpassmask = ones(M,N); mybandpassmask =  
ones(M,N);  
for u = 1:M for  
v = 1:N  
    tmp = ((u-(M+1))/2)^2 + (v-(N+1)/2)^2; raddist =  
round(sqrt(tmp)); disp(raddist) if raddist > low  
mylowpassmask(u,v) = 0; end  
if raddist > band2 || raddist < band1;  
mybandpassmask(u,v) = 0; end end end  
f1 = fftshift(mylowpassmask);  
f3 = fftshift(mybandpassmask);
```

```
resimage1 = myfftimage.*f1; resimage3 =  
myfftimage.*f3;
```

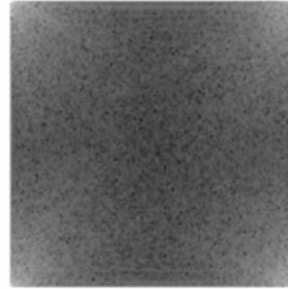
```
r1 = abs(fft2(resimage1)); subplot(2,2,3);  
imshow(r1,[]),title('Low Pass filtered image');
```

```
r3 = abs(fft2(resimage3)); subplot(2,2,4);  
imshow(r3,[]),title('Band Pass filtered image');
```

Original Image



FFT Image



Low Pass filtered image



Band Pass filtered image



EXPERIMENT NO: 5

Objective: Perform various Morphological operation dilation, erosion, internal & external boundary Extraction; Thinning, Thickening & Perform Dilation, erosion, boundary Extraction using direct function.

1. Dilation image

Erosion image

Internal Boundary Extraction

External Boundary Extraction

Morphological gradient

Thinning of the image

Thickening of the image

Skeletonization of image

2. perform Dilation, erosion, boundary Extraction without using direct function

Perform various Morphological operation on their applications

```
myorigimg = imread('text.png');
subplot(3, 3, 1);
imshow(myorigimg);title('Originalimage');

se = strel('disk', 9);

mydilatedimg = imdilate(myorigimg, se); subplot(3, 3, 2);
imshow(mydilatedimg);title('Dilated image');

myerodedimg = imerode(myorigimg, se); subplot(3, 3, 3);
imshow(myerodedimg);title('Eroded image');

internalboundimg = mydilatedimg & ~ myerodedimg; subplot(3, 3, 4);
imshow(internalboundimg,[]);title('Internal Boundary');

externalboundimg = mydilatedimg & ~myorigimg; subplot(3, 3, 5);
imshow(externalboundimg,[]);title('External Boundary');

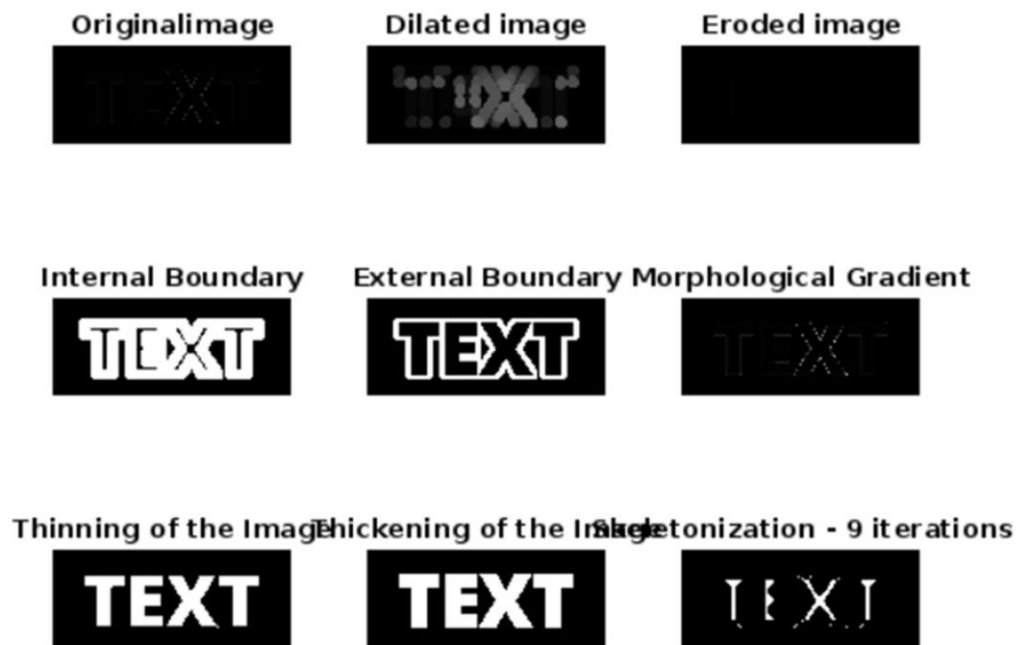
mymorphgradimg = imsubtract(myorigimg,myerodedimg); subplot(3, 3, 6);
imshow(mymorphgradimg,[]);title('Morphological Gradient');

thin = bwmorph(myorigimg,'thin'); subplot(3,3,7);
imshow(thin);title('Thinning of the Image');

thickf = bwmorph(myorigimg,'thicken'); subplot(3,3,8);
imshow(thickf);title('Thickening of the Image');

skelf100 = bwmorph(myorigimg,'skel',9); subplot(3,3,9);
```

```
imshow(skelf100);title('Skeletonization - 9 iterations');
```



perform Dilation, erosion, boundary Extraction without using direct function

```
a=imread('dog.jfif'); p=size(a);
w=[1 1 1; 1 1 1; 1 1 1]; for
x=2:1:p(1)-1 for y=2:1:p(2)-1
    a1=[w(1)*a(x-1,y-1) w(2)*a(x-1,y) w(3)*a(x-1,y+1) w(4)*a(x,y-1) w(5)*a(x,y) w(6)*a(x,y+1)
w(7)*a(x+1,y-1) w(8)*a(x+1,y) w(9)*a(x+1,y+1)];    A(x,y)=min(a1);%Erosion
    B(x,y)=max(a1);%dilation
    Sharp(x,y)=a(x,y)-A(x,y); end end
subplot(2,2,1),imshow(a),title('original image'); subplot(2,2,2),imshow(A),title('erosion');
subplot(2,2,3),imshow(B),title('Dilation');
subplot(2,2,4),imshow(Sharp),title('boundary extracted');
```

original image



erosion



Dilation



boundary extracted



EXPERIMENT NO : 6

Objective: Perform image segmentation.

```
img1 = imread('download (4).jpg'); img2 =  
rgb2gray(img1); [m,n] = size(img2); img3 =  
zeros(m,n); for i=1:m    for j=1:n  
if(img2(i,j)<150)        img3(i,j)=0;  
else                    img3(i,j)=1;    end    end end  
subplot(1,2,1), imshow(img1);  
subplot(1,2,2), imshow(img3);
```



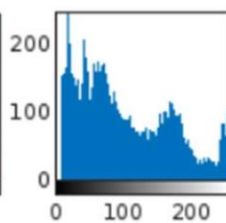
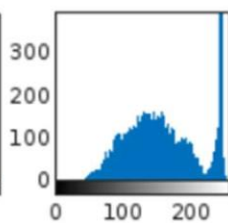
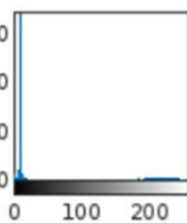
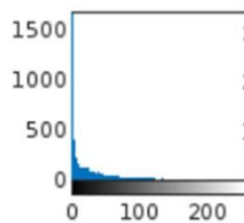
EXPERIMENT NO : 7

Objective: Perform histogram equalization.

```
img1=imread('dark.jpg');
img2=rgb2gray(img1); imhist(img2);
img9=histeq(img2);

img3=imread('high contrast image.jpg');
img4=rgb2gray(img3); imhist(img4);
img10=histeq(img4);

img5=imread('low contrast image.jpg');
img6=rgb2gray(img5); imhist(img6);
img11=histeq(img6);
img7=imread('lightimage.jpg');
img8=rgb2gray(img7); imhist(img8);
img12=histeq(img8);
subplot(3,4,1), imshow(img1)
subplot(3,4,2), imshow(img3)
subplot(3,4,3), imshow(img5)
subplot(3,4,4), imshow(img7)
subplot(3,4,5), imhist(img2)
subplot(3,4,6), imhist(img4)
subplot(3,4,7), imhist(img6)
subplot(3,4,8), imhist(img8)
subplot(3,4,9), imshow(img9)
subplot(3,4,10), imshow(img10)
subplot(3,4,11), imshow(img11)
subplot(3,4,12), imshow(img12)
```



EXPERIMENT NO: 8

The objective of this lab session is to understand Segmentation and implement the following tasks in MATLAB

Simple thresholding

Multiple thresholding

Adaptive thresholding

Optimal thresholding

```
a = imread('rose.jpeg'); a =  
rgb2gray(a); subplot(3,3,1);  
imshow(a); title('Original Image');
```

```
level = 0.3;
```

```
subplot(3,3,2); segimage1 =  
im2bw(a,level);  
imshow(segimage1); title('Simple Thresholding at 0.3');
```

```
subplot(3,3,3);  
imshow(a > 153); title('Simple Thresholding at 0.6');  
    tmp = a; [m n]=  
find(a<26); for j = 1:  
length(m)  
tmp(m(j),n(j))=0; end  
[m n]= find(a>26 & a <= 230); for j =  
1: length(m)    tmp(m(j),n(j))=0.8; end  
[m n]= find(a>230); for j = 1:  
length(m)    tmp(m(j),n(j))=0;  
end subplot(3,3,4); segimage2 =  
im2bw(tmp,0);  
imshow(segimage2); title('Multiple thresholding(Between 27-230)');
```

```
level = graythresh(a);
```

```
subplot(3,3,5); segimage =  
im2bw(a,level);  
imshow(segimage); title('Otsu - Optimal Segmented Image');
```

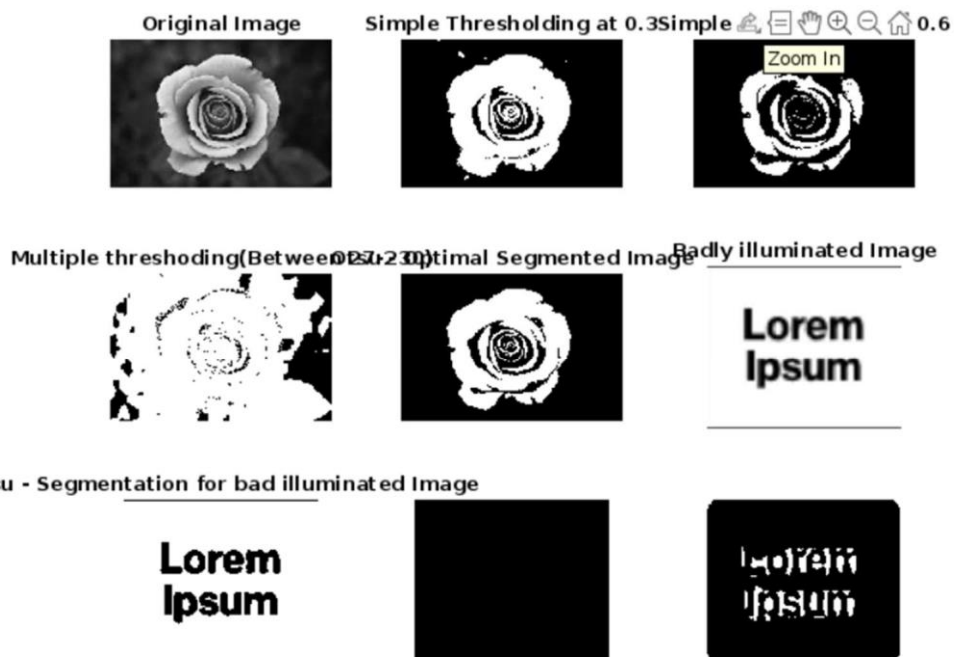
```
b = imread('blurred.jpeg'); subplot(3,3,6);  
imshow(b); title('Badly illuminated Image'); level =  
graythresh(b); subplot(3,3,7); segimage = im2bw(b,level);  
imshow(segimage); title('Otsu - Segmentation for bad illuminated Image'); b =  
imread('blurred.jpeg'); b = rgb2gray(b);
```

```
avgfilt = ones(13,13);
adaptfiltmask = avgfilt/sum(avgfilt); im =
imfilter(b,adaptfiltmask,'replicate');
```

```
im1 = medfilt2(b,[20 20]);
```

```
thresh = im+18; adaptthreshimg = b -
thresh; subplot(3,3,8);
imshow(adaptthreshimg > 0);
```

```
thresh1 = im1 + 2; adaptthreshimg = b -
thresh1; subplot(3,3,9);
imshow(adaptthreshimg > 0);
```



EXPERIMENT NO: 9

The objective of this lab session is to understand edge detection operators in MATLAB.

Ordinary operator

Roberts's operator

Prewitts operator

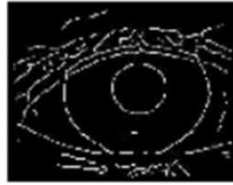
Sobel operator

```
a=imread('sovel.jpeg');
[row col]=size(a);

w1=[1 0; -1 0]; w2=[1 -1; 0
0]; w3=[1 0; 0 -1]; w4=[0 1;
-1 0];

w5=[-1 0 -1;-1 0 1;-1 0 1]; w6=[-1 -1 -1;0 0
0;1 1 1];

w7=[-1 -2 -1;0 0 0; 1 2 1]; w8=[-1 0
1;-2 0 2; -1 0 1]; for x=2:row-1
for y=2:col-1
    a1(x,y)=w1(1)*a(x,y)+w1(2)*a(x,y+1)+w1(3)*a(x+1,y)+w1(4)*a(x+1,y+1);
a2(x,y)=w2(1)*a(x,y)+w2(2)*a(x,y+1)+w2(3)*a(x+1,y)+w2(4)*a(x+1,y+1);
a3(x,y)=w3(1)*a(x,y)+w3(2)*a(x,y+1)+w3(3)*a(x+1,y)+w3(4)*a(x+1,y+1);
a4(x,y)=w4(1)*a(x,y)+w4(2)*a(x,y+1)+w4(3)*a(x+1,y)+w4(4)*a(x+1,y+1);    a5(x,y)=w5(1)*a(x-
1,y-1)+w5(2)*a(x-1,y)+w5(3)*a(x-1,y+1)+w5(4)*a(x,y-
1)+w5(5)*a(x,y)+w5(6)*a(x,y+1)+w5(7)*a(x+1,y1)+w5(8)*a(x+1,y)+w5(9)*a(x+1,y+1);
    a6(x,y)=w6(1)*a(x-1,y-1)+w6(2)*a(x-1,y)+w6(3)*a(x-1,y+1)+w6(4)*a(x,y-
1)+w6(5)*a(x,y)+w6(6)*a(x,y+1)+w6(7)*a(x+1,y1)+w6(8)*a(x+1,y)+w6(9)*a(x+1,y+1);
    a7(x,y)=w7(1)*a(x-1,y-1)+w7(2)*a(x-1,y)+w7(3)*a(x-1,y+1)+w7(4)*a(x,y-
1)+w7(5)*a(x,y)+w7(6)*a(x,y+1)+w7(7)*a(x+1,y1)+w7(8)*a(x+1,y)+w7(9)*a(x+1,y+1);
    a8(x,y)=w8(1)*a(x-1,y-1)+w8(2)*a(x-1,y)+w8(3)*a(x-1,y+1)+w8(4)*a(x,y-
1)+w8(5)*a(x,y)+w8(6)*a(x,y+1)+w8(7)*a(x+1,y1)+w8(8)*
a(x+1,y)+w8(9)*a(x+1,y+1);    end end A1=a1+a2;
A2=a3+a4;
A3=a5+a6; A4=a7+a8;
figure(1),subplot(2,3,1),imshow(a),title('original image');
figure(1),subplot(2,3,2),imshow(uint8(A1)),title('ordinary operator');
figure(1),subplot(2,3,3),imshow(uint8(A2)),title('Roberts Operator');
figure(1),subplot(2,3,4),imshow(uint8(A3)),title('Prewitts Operator');
figure(1),subplot(2,3,5),imshow(uint8(A4)),title('Sobel Operatos');
```



ordinary operator



Robert's Operator



Prewitt's Operator



Sobel Operator



EXPERIMENT NO: 10

Perform dilation, erosion, opening and closing operation on image and consider $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ as structural element.

```
a=imread('dog.jfif'); p=size(a);  
w=[1 1 1;1 1 1;1 1 1];  
  
for x=2:1:p(1)-1    for  
y=2:1:p(2)-1  
    a1=[w(1)*a(x-1,y-1) w(2)*a(x-1,y) w(3)*a(x-1,y+1) w(4)*a(x,y-1) w(5)*a(x,y) w(6)*a(x,y+1)  
w(7)*a(x+1,y-1) w(8)*a(x+1,y) w(9)*a(x+1,y+1)];  
    A(x,y)=min(a1);%Erosion  
    B(x,y)=max(a1);%dilation  
Sharp(x,y)=a(x,y)-A(x,y);    end end  
subplot(2,2,1),imshow(a),title('Original Images'); subplot(2,2,2),imshow(A),title('Erosion Images');  
subplot(2,2,3),imshow(B),title('Dilation Images'); subplot(2,2,4),imshow(Sharp),title('Boundar Images');
```

Original Images



Erosion Images



Dilation Images



Boundar Images

